

# Loss Function Optimization for TCN-based Model For Surgical Phase Recognition

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**Abstract**—Surgical phase recognition from video data is crucial for enhancing medical procedures and training. While binary cross-entropy is a standard loss function for classification, it often overlooks the inherent temporal consistency of surgical workflows. This work proposes an augmented loss function to improve the performance of Temporal Convolutional Network (TCN)-based models for surgical phase recognition. Our approach integrates three novel components: a transition loss to penalize biologically implausible phase transitions, a smoothness loss to reduce noisy, abrupt changes in predictions, and a multi-task loss that leverages tool predictions as an auxiliary task. We apply this augmented loss function to the TeCNO architecture and evaluate it on the public Cholec80 dataset, comprising 80 cholecystectomy surgeries. Our experiments demonstrate that the multi-task and smoothness losses yield improvements in accuracy, precision, and recall compared to the standard binary cross-entropy. The transition loss, however, did not show significant improvement, likely due to TeCNO's inherent temporal awareness. These findings suggest that incorporating domain-specific temporal characteristics and auxiliary tasks into the loss function can enhance surgical phase recognition models, with varying degrees of impact depending on the specific loss component.

**Index Terms**—Surgical phase recognition, Temporal CNN, Causal Convolution, Temporal Consistency, Loss Function Optimization, Deep Learning for Medical Applications, cholec80

## I. INTRODUCTION

THE classification of surgical phases from video data has lately been a subject of attention in the medical field, possibly due to its potential in enhancing medical procedures and data acquisition, which can later be used for training and/or evaluations. The continuous rise of new ideas and high-performing Deep Learning models has aided that cause. In such models, the most common loss function encountered is binary cross-entropy, which is a standard approach for classification problems. In this work, we present an alternative, augmented loss function which takes advantage of the inherent characteristic of "temporal consistency" of a surgery's phases, along with a multi-task loss that utilizes tool predictions. Finally, we added a smoothness loss to penalize noisy transitions. The first method takes advantage of the fact that surgical phases have a certain logical and temporal flow apart from the visual information that the model learns to relate with the known phases. The second method, utilizes an extra output head of the network, which predicts tools. Using that extra information, the network performs better in the main objective, which is phase predictions. To prove that models can benefit from our custom loss function, we applied it to an architecture named TeCNO- Surgical Phase Recognition with Multi-Stage

Temporal Convolutional Networks [1], using the cholec80 public dataset which consists of 80 videos of cholecystectomy surgeries performed by 13 surgeons.

In the Methodology section, we describe how the cholec80 dataset was split for our experiments. We then present how we implemented the augmented loss functions. These include: A Multi-task version of the network that utilizes tool predictions, a transition loss based on an adjacency matrix of the surgical phases, which enforces the logical sequence of the phases, and finally a smoothness loss to penalize high frequency predictions. In the Experiments & Results we present a table with the average metric scores and compare them with the scores of the original model. We prove that our custom loss function is superior to the standard BCE loss, for the specific task of surgical phase recognition, a conclusion which can possibly be extended to any model that classifies a procedure's phases in general.

## II. RELATED WORK

The task of surgical phase recognition from intra-operative videos is challenging due to several critical factors: variability in patient anatomy, diverging surgeon styles, and limited availability and quality of annotated video data [1] [2]. Traditional temporal models such as RNNs, LSTMs, and Transformers have been commonly employed, yet each exhibits distinct limitations.

In the current literature, we may find mostly RNNs (Recurrent Neural Networks), LSTMs and Transformers to perform this task. LSTMs can capture sequential dependencies but are inherently constrained by their short-to-mid-range memory. Considering surgeries often last minutes to hours, this limited receptive length hinders effective long-term context modeling [2].

To enhance LSTM-based models, Jin et al. [2] introduced a multi-task framework for tool detection and phase recognition. A novel correlation loss penalizes divergence between tool presence and phase prediction, embedding domain-specific intuition: tools co-occur with phases clinically. On the Cholec80 dataset, this boosted tool detection mAP from 81.0% to 89.1%, and phase recognition F1 from 84.5% to 87.4%. Our addition of a multi-task head for loss function augmentation was inspired by their work.

Temporal Convolutional Networks (TCNs) were first applied to general action segmentation to hierarchically process visual sequences. By combining encoder-decoder structures with dilated convolutions, TCNs increased the receptive field

while keeping computational costs low, with capabilities beyond RNNs [3] [4].

An interesting idea to improve the standard TCN architecture was proposed by Fahra et al. [5], who employed a multi-stage TCN network with decoupled refinement stages and a dual-dilated convolutional layer that uses both large and small receptive fields. They also employed a custom smoothness loss to prevent over-segmentation errors. To implement that loss, they used a truncated mean squared error over the frame-wise log-probabilities.

Our work focuses on improving the TeCNO model, which was the first to introduce a multi-stage TCN approach with causal dilated convolutions for surgical phase recognition [1]. The TeCNO model can be viewed in Figure 1. TeCNO adapts the MS-TCN architecture for causal and online surgical phase recognition.

The process of TeCNO model starts with ResNet-50 to extract per-frame spatial features. Then introduces a 2-stage causal TCN that processes only preceding frames, ensuring real-time applicability. Dilated convolutions enable a wide temporal context (spanning several minutes) and yield smooth, consistent predictions, especially during ambiguous transitions.

At the end it is evaluated on laparoscopic cholecystectomy datasets.

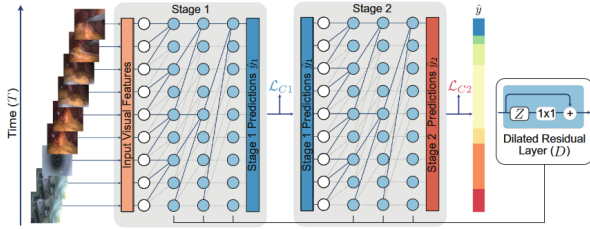


Fig. 1: Overview of the TeCNO multi-stage hierarchical refinement model

### III. METHODOLOGY

#### A. Data

The Cholec80 dataset comprises 80 laparoscopic cholecystectomy videos performed by 13 different surgeons [6]. These videos are recorded at a frame rate of 25 frames per second (fps). Each frame is annotated with surgical phase labels at 25 fps, while tool presence is annotated at a lower frequency of 1 fps. The surgical phases were defined by an experienced senior surgeon. To account for the challenges in visually detecting surgical instruments particularly when tools are only partially visible a tool is considered present in a frame if at least half of its tip is visible.

#### B. Dataset Split

Our approach and the paper, intentionally uses a fixed hold-out split instead of cross-validation.

- 40 videos for training (videos 1-40)
- 8 videos for validation (videos 41-48)
- 32 videos for testing (videos 49-80)

#### C. Idea

Combining TeCNO's long-range causal context with three different kind of loss functions. Such as transition, Smoothness and Multi-Task Loss.

#### D. Feature Extractor

The feature extractor works with a CNN that predict the phase of a surgery (multiclass classification). TeCNO introduces a second output head which performs tool predictions and gets trained based on their accuracy. This way, the stem of the model, which essentially comprises of the layers before the two-headed output, gets trained so it performs well both on tools and on phases (multilabel classification). It is important to note however, that this first stage is only a rough prediction of the phases, since it utilizes only visual information. ResNet50 outputs 3 things:

- stem: a feature vector
- phase: logits for phase classification
- tool: logits for tool classification (if task is enabled)

It gets trained using two loss functions:

- Cross-entropy for phase prediction
- Binary cross-entropy for tool prediction

Losses are automatically balanced using learned log-variance weights. It tracks accuracy and F1-score during training and validation. It also logs and saves predictions per video, including: Raw model outputs (stem), Predicted and ground truth labels. Finally, one .pkl file containing all the features extracted is exported for each video. The file contains

- 1) The extracted visual features
- 2) The initial phase predictions from the CNN
- 3) The ground truth phase labels
- 4) The ground truth tool presence labels (Only for the multi-task implementation, where tool logits are required).

#### E. The Multi-Stage Temporal Convolutional Network (MSTCN)

The stages that follow after ResNet50 are 2 TCN models, which are the innovative parts of the TeCNO architecture that enhance the predictions of the phases. These models were devised in order to achieve temporal awareness by separating two concerns :

- 1) Using the visual features to deduct the current phase
- 2) Creating a temporal relation between the phases that are being predicted.

TeCNO achieves this with the 2-staged TCN. The first model receives as inputs only the visual features, and learns to relate them to a phase prediction, while the second stage receives these initial predictions and improves them by learning to put them in a temporal context.

Our implementations of custom loss functions lead us into working on two different model architectures:

- 1) TeCNO, the model along with the feature extraction steps as was provided from the authors of TeCNO, with minor adjustments on library versioning and our custom loss functions for smoothness loss and transition loss.

- 2) Multi-Task version of TeCNO, which introduces tool information in the feature extractor, and an extra output head related to tool predictions. Additionally, it contains the extra code required for the multi-task loss implementation. A weighted BCE-loss was used for the tool prediction, identically with the original phase predictions.

#### F. Custom Loss Functions

Several customized loss function were tried and by the end we stuck with three of them:

- Transition Loss
- Smoothness Loss
- Multi-Task Loss

We will exploit them in detail in the paragraphs below.

1) *Transition Loss*: Inspired by findings from the literature, a transition loss was implemented to enforce temporal consistency and penalize biologically implausible phase transitions during surgical workflow modeling.

The transition loss operates based on the following procedure:

- It analyzes all transitions between surgical phases in the training data.
- It counts the frequency and directionality of these transitions.
- It constructs a binary transition matrix indicating which phase-to-phase transitions are allowed.

The core idea is straightforward: If a model predicts a transition that is valid according to the learned transition matrix (e.g., staying in the same phase or moving from Phase 0 to Phase 1), no penalty is applied. However, if the model predicts an invalid transition (e.g., jumping from Phase 0 directly to Phase 5), it incurs a penalty.

To make this penalty more expressive, a distance-based weighting is introduced. Transitions to distant or highly implausible phases are penalized more severely than those to nearby or adjacent phases in the typical surgical workflow. For instance, confusing the "Preparation" phase with "Cleaning" results in a higher penalty than confusing it with "Calot Triangle Dissection."

An implementation detail arose during training: Initially, the model predictions were reduced using argmax to determine the predicted phase class. However, since argmax is non-differentiable, it prevented proper gradient flow and model learning. This issue was resolved by replacing argmax with softmax outputs, which are differentiable and allow the model to learn from the transition loss gradients effectively.

2) *Smoothness Loss*: Surgical phases in videos typically change slowly and follow a logical flow. However, deep learning models can sometimes produce predictions that jump abruptly between phases in very short time spans. These fast and noisy switches are mostly undesirable and reduce the reliability of the model.

To prevent this, we add a smoothness loss to our total loss function. This loss penalizes sudden and incorrect changes in the predicted labels frame by frame, so the model is encouraged to produce more stable and realistic predictions.

This total loss combined with smoothness loss is calculated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{smooth}}$$

where  $\mathcal{L}_{\text{CE}}$  is the classification loss, computed using the standard cross-entropy and  $\mathcal{L}_{\text{smooth}}$  is the smoothness loss, which penalizes abrupt or incorrect changes in predicted phase labels. Also  $\lambda$  as a weighting factor that balances the contribution of the smoothness loss.

The smoothness loss works by checking each pair of consecutive frames:

- If the ground-truth label stays the same between frame  $t$  and  $t + 1$ :
  - And the model changes its prediction to a wrong class  $\Rightarrow$  penalize.
  - And the model changes to the correct class  $\Rightarrow$  no penalty.
  - And the prediction does not change  $\Rightarrow$  no penalty.
- If the ground-truth label changes between  $t$  and  $t + 1$ :
  - No penalty is applied (even if prediction changes or not).

This custom smoothness loss ensures that the model only gets penalty when it makes unnecessary or incorrect prediction switches in stable regions. As a result, the model learns to make smoother and more temporally consistent decisions.

3) *Multi-Task Loss*: Multi-task networks have the characteristic of learning different tasks, in order to enhance the model's capability to predict the main task. Due to implicit relations between different types of data, this methodology has been proven beneficial to the overall predictions of models.

In our implementation, the network was adapted to concurrently predict both the primary surgical phase and, as an auxiliary task, the set of tools present in each frame. This was achieved by modifying the architecture of each stage in the temporal network to include a secondary output head, dedicated to tool prediction, which operates in parallel to the original phase prediction head. The core dilated convolutional layers that learn the temporal features of the video remain unchanged, but are now tasked with creating a shared representation that is rich enough to inform both prediction tasks. Consequently, the model was trained on a loss function which consists of a weighted sum of the multi-class cross-entropy loss from the phase recognition task and the binary cross-entropy loss from the multi-label tool prediction task. This approach encourages the model to learn a more robust and context-aware feature representation, using the explicit tool supervision to improve its primary goal of surgical phase recognition. The weighing factor of the tool predictions proved most beneficial at low values, close to 0.01, suggesting that an overfitting of the model to the secondary objective can occur if a large factor is selected.

## IV. EXPERIMENTS AND RESULTS

The aim of our experiments is to prove that our custom loss function caused an improvement in three metric scores: Accuracy, Precision and Recall. We have used macro-averaging

of the metric scores, which means we compute the metrics independently for each class, then take the unweighted average across classes. The results reported in this section include all attempted methods : Smoothness Loss, Transition Loss, Correlation Loss, Multi-task loss. Not all of these methods yielded increased metric scores, so the results will be later discussed.

The hyperparameters used were the same for each experiment, the number of epochs varies and it stops automatically when the validation result don't improve for 3 consecutive epochs.

TABLE I: Comparison of Multi-task loss with basic loss (mean and std)

| Method                 | Accuracy          | Precision         | Recall            |
|------------------------|-------------------|-------------------|-------------------|
| <b>Multi-task loss</b> | $0.855 \pm 0.012$ | $0.783 \pm 0.020$ | $0.830 \pm 0.006$ |
| <b>Basic Loss</b>      | $0.851 \pm 0.014$ | $0.774 \pm 0.022$ | $0.827 \pm 0.009$ |
| Improvement(%)         | 0.37              | 0.95              | 0.24              |

TABLE II: Comparison of Smoothness loss with basic loss (mean and std)

| Method                 | Accuracy          | Precision         | Recall            |
|------------------------|-------------------|-------------------|-------------------|
| <b>Smoothness Loss</b> | $0.891 \pm 0.006$ | $0.797 \pm 0.009$ | $0.824 \pm 0.009$ |
| <b>Basic Loss</b>      | $0.885 \pm 0.008$ | $0.796 \pm 0.012$ | $0.822 \pm 0.009$ |
| Improvement(%)         | 0.69              | 0.05              | 0.27              |

TABLE III: Comparison of Transition loss with basic loss (mean and std)

| Method                 | Accuracy          | Precision         | Recall            |
|------------------------|-------------------|-------------------|-------------------|
| <b>Transition Loss</b> | $0.891 \pm 0.005$ | $0.799 \pm 0.008$ | $0.818 \pm 0.007$ |
| <b>Basic Loss</b>      | $0.891 \pm 0.004$ | $0.801 \pm 0.009$ | $0.818 \pm 0.010$ |
| Improvement(%)         | 0.00              | -0.28             | 0.00              |

## V. CONCLUSION

In this project, we extended the TeCNO architecture for surgical phase recognition and we developed and examined a set of custom loss functions to improve the model performance.

Specifically, we implemented and evaluated three losses: a transition loss to discourage impossible phase jumps, a smoothness loss to penalize unnecessary phase changes, and a multi-task loss that uses the tool presence as a secondary task.

Our method takes advantage of the fact that surgical videos change over time in a predictable way. We designed the model to make smoother and more logical predictions across frames. We also added multi-task learning, which helps the model learn better by using extra information about which tools are used during each phase.

The results show that each custom loss function had a different impact. The multi-task loss provided a small improvement across all metrics, showing that using tool information can be helpful, though the gain is limited. The smoothness loss had a slightly bigger effect on accuracy, with minor improvements on recall and precision as well. And for the transition loss, it did not make any improvement in accuracy and it reduced precision a little. We attribute this result to the model's

inherent capability to have good temporal awareness using the two-staged TCN implementation, which the transition loss probably did not enforce.

Our results showed that while some changes to the loss function can help improve model performance, not all applied methods have the positive impact. Tuning of the regularization factors is also very important, because when too much weight is given to the secondary loss terms, the model tends to overfit the secondary objective and lose overall performance.

## ACKNOWLEDGMENT

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